

Divam Shah

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🌐 Codeforces | 📍 Surat, Gujarat

Education

Sardar Vallabhbhai National Institute of Technology, Surat
B.Tech in Artificial Intelligence / CPI: 8.35/10

2024–2028
Surat, Gujarat

– Relevant Coursework: Data Structures, DBMS, Operating Systems, Computer Networks, Artificial Intelligence

Technical Skills

Programming Languages: C++, JavaScript, Python, SQL

Web Technologies: React, Next.js, Node.js, Express.js, Socket.io, Tailwind CSS

AI/ML & Data: RAG, Vector Databases, pgvector, Gemini API, XGBoost, PyMuPDF

Databases & Cloud: PostgreSQL, MongoDB, Supabase, Cloudinary API, REST APIs

Tools & Practices: Git, GitHub, Postman, JWT/OAuth, Agile Methodologies

Projects

ChatPDF – AI-Powered Document Assistant | Next.js, FastAPI, Gemini, Supabase, pgvector

- Built a full-stack Retrieval-Augmented Generation (RAG) platform that enables users to upload PDFs and ask natural language questions with grounded, source-backed answers.
- Developed an end-to-end retrieval pipeline using PyMuPDF, Gemini Embeddings, pgvector, and Supabase Storage to perform semantic search over document chunks.
- Implemented citation-aware responses with clickable references that navigate directly to the corresponding PDF page and highlight the cited text using bounding-box overlays.
- Designed persistent conversational memory and per-user document isolation using JWT authentication, enabling secure multi-user interactions.
- Engineered robust retry mechanisms and exponential backoff strategies for Gemini API rate limits, significantly improving system reliability on free-tier infrastructure.
- Deployed the production-ready application using Vercel, Render, MongoDB Atlas, and Supabase with automated document processing and vector indexing workflows.

Algorithmic Trading Bot

- Code
- Built a deployment-ready algorithmic trading platform using Next.js and FastAPI microservices architecture.
- Implemented Hidden Markov Models, Kalman Filter pairs trading, and Ornstein-Uhlenbeck mean reversion strategies.
- Developed a backtesting engine with Kelly Criterion position sizing, stop-loss controls, and volatility risk management.
- Designed an interactive frontend with real-time charts and REST API integration.

Achievements

Codeforces: Max Rating: 1400+

Second Runner-Up – Powermind Hackathon hosted by Adani and CDC NIT Surat.

Experience

Mindbend, SVNIT Surat

Co-Head – Sponsorship Team

- Led sponsorship outreach and collaboration efforts for the institute’s flagship technical festival.
- Coordinated with industry partners and managed sponsor communications, negotiations, and deliverables.
- Contributed to sponsorship planning and execution strategies to support event funding and partnerships.

Google Developer Groups on Campus (GDGC)

Junior Developer

- Contributed to campus developer initiatives, technical events, and collaborative software projects.
- Worked with peers on modern development practices, software development workflows, and community learning activities.

Certifications

Google Cloud Study Jams – Completed guided labs on cloud infrastructure, storage, networking, deployment, and DevOps tools using Google Cloud Platform.